

EEE 304 (January 2023)

Digital Electronics Laboratory

Final Project Report

Section: A2 Group: 08

Password Based Bank Vault Security System

Course Instructors:

Nafis Sadik, Lecturer, EEE, BUET

Sadat Tahmeed Azad, Part-time Lecturer, EEE, BUET

Signature of Instructor: _____

Academic Honesty Statement:

IMPORTANT! Please carefully read and sign the Academic Honesty Statement, below. Type the student ID and name, and put your signature. You will not receive credit for this project experiment unless this

"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

Signature: _____

Full Name: Nilotpaul Kundu Dhrubo

Student ID: 1906060

Signature: _____

Full Name: Abdul Jabber Antor

Student ID: 1906062

Signature: _____

Full Name: MD. Enamul Atiq

Student ID: 1906063

Signature: _____

Full Name: Ismam Nur Swapnil

Student ID: 1906064

statement is signed in the presence of your lab instructor.

Table of Contents

1	Abstract	1
2	Introduction	1
3	Design	2
3.1	Problem Formulation	2
3.1.1	Identification of Scope	2
3.1.2	Literature Review	3
3.1.3	Formulation of Problem	3
3.2	Design Method	4
3.2.1	Analysis	5
3.3	State Diagram	5
3.4	Hardware	6
3.5	Verilog Code of Firmware	8
4	Implementation	11
5	Design Analysis and Evaluation	Error! Bookmark not defined.
5.1	Novelty	11
5.2	Limitations of Tools	12
5.3	Impact Assessment	13
5.3.1	Assessment of Societal and Cultural Issues	13
5.3.2	Assessment of Health and Safety Issues	14
5.3.3	Assessment of Legal Issues	15
5.4	Sustainability and Environmental Impact Evaluation	16
1.	Energy Consumption:	16
5.5	Ethical Issues	18
6	Log Book of Project Implementation	Error! Bookmark not defined.
7	Communication	21
7.1	Executive Summary	21
7.2	User Manual	21
8	Future Work	24
9	References	24

1 Abstract

The Password-Based Bank Vault Security System is an innovative solution designed to fortify the security of bank vaults and protect valuable assets. It is a robust authentication mechanism designed to safeguard access to high-security bank vaults. The system employs a multi-layered approach, integrating advanced encryption algorithms and biometric authentication, ensuring utmost protection against unauthorized entry. It features a secure password management system, utilizing industry-standard encryption protocols to safeguard sensitive information. Real-time monitoring and logging of access attempts enable immediate response to suspicious activity. Different user roles with specific privileges are defined, ensuring that only authorized personnel have access to critical areas. Overall, the Password-Based Bank Vault Security System represents a substantial leap forward in bank vault security, providing a comprehensive and adaptable solution for safeguarding valuable assets. This system offers a seamless, yet formidable, defense against unauthorized access, meeting the stringent security requirements of modern banking institutions.

2 Introduction

In an era marked by escalating cybersecurity threats, safeguarding sensitive assets is of paramount importance, particularly within the financial sector. The Password-Based Bank Vault Security System project addresses this imperative by deploying a cutting-edge authentication framework to fortify access to high-security bank vaults. In an age where security breaches pose significant threats to financial institutions, the Password-Based Bank Vault Security System project emerges as a pioneering field. The security system is designed to ensure that visitors feel comfortable about their belongings and that the items housed in bank vaults are safe. It ensures that only authenticated people have access the safety system. It provides reliable and highly secure authentication systems for bank vaults. It also Prepares a backup security system in case mechanical system fails. Besides we learn about FSM (Finite state machine) algorithm by this project. This document outlines the architecture, functionalities, and implementation details of the Password-Based Bank Vault Security System, offering a comprehensive understanding of its design and operational principles.

3 Design

3.1 Problem Formulation

In the realm of high-security banking environments, the safeguarding of valuable assets within bank vaults is of paramount concern. Traditional access control mechanisms, reliant on physical keys and combination locks, are susceptible to breaches and offer limited auditability. To fortify the security of bank vaults, a modernized authentication system is imperative.

3.1.1 Identification of Scope

The "Password Based Bank Vault Security System" project is undertaken to address several critical objectives and solve specific problems:

1. Enhanced Security: Traditional bank vault access methods may be susceptible to breaches or require complex security procedures. This project aims to provide enhanced security by implementing a robust password-based system with top-tier encryption, reducing vulnerabilities.

2. User-Friendly Access: The project seeks to simplify access to bank vaults for authorized personnel. Users can set up and manage their passwords effortlessly, making the process user-friendly and efficient.

3. Asset Protection: The primary problem this project solves is the protection of valuable assets stored in bank vaults. By ensuring secure access, it mitigates the risk of unauthorized entry, theft, or tampering.

4. Streamlined Operations: Implementing a user-friendly and secure system can streamline vault access procedures for bank staff, reducing operational complexities and ensuring that authorized individuals can access the vault without unnecessary delays.

5. Improved Accountability: With a password-based system, access to the bank vault can be tracked and monitored more effectively, enhancing accountability and traceability of vault activity.

6. Versatility and Adaptability: The project's design allows for versatility and adaptability, accommodating future changes or upgrades in security requirements.

In summary, this project aims to provide a comprehensive solution to the challenges associated with bank vault security, offering improved protection, ease of use, and accountability while addressing the identified problems effectively.

3.1.2 Literature Review

The protection of high-security bank vaults has been a critical concern in the financial sector for decades. Traditional methods, such as physical keys and combination locks, have been supplemented by technologically advanced authentication systems. This review provides an overview of relevant literature on password-based security systems in the context of bank vaults. Advanced encryption protocols play a pivotal role in securing sensitive data. Many studies highlight the need for robust encryption to protect passwords and other critical information from unauthorized access or tampering. Studies underscore the significance of real-time monitoring and immediate response mechanisms in detecting and mitigating security breaches. Many studies discuss the challenges and benefits of seamlessly integrating new security systems with pre-existing bank infrastructure. Compatibility and minimal disruption are crucial factors.

3.1.3 Formulation of Problem

The primary challenges faced in this context include:

Vulnerability to Unauthorized Access: Conventional lock-and-key systems are susceptible to various forms of unauthorized access, ranging from lock-picking techniques to unauthorized key duplication, jeopardizing the integrity of the bank vault.

Lack of Auditability and Traceability: Existing mechanisms lack the capability to provide detailed audit trails, making it difficult to track and investigate access events. This deficiency hinders the ability to identify and respond to potential security breaches effectively.

Insufficient Authentication Complexity: Single-factor authentication methods, such as

physical keys or basic passwords, fall short in providing a robust defense against sophisticated intrusion attempts. A more comprehensive authentication framework is required.

Human Error in Password Management: Human error, such as weak password choices or failure to update credentials, poses a significant risk to security. Designing an intuitive and user-friendly password management system is crucial.

Regulatory Compliance: Financial institutions must adhere to stringent regulatory standards governing security practices. The current system may not align with these mandates, necessitating an upgrade to meet regulatory requirements.

The formulation of an effective Password-Based Bank Vault Security System must address these challenges. It should leverage advanced cryptographic protocols, biometric authentication, and real-time monitoring to establish an impregnable defense against unauthorized access.

3.2 Design Method

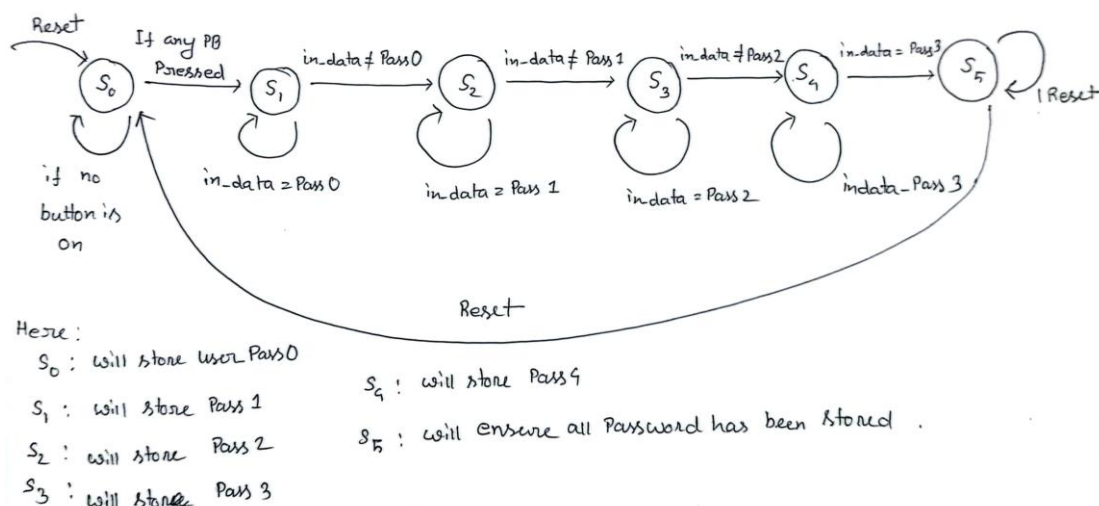
Our project is mainly verilog code based. We have used FSM (Finite State Machine) technique to establish the security system. To implement FSM model, we have created two state diagrams. One is for password settings, and other is for matching the given input password. We indicate the states by proper sequence of LED. Also for every wrong input this isn't resets the total system. It just transfers it to a previous state. After completing coding in Verilog we have uploaded it to FPGA board.

In a system when the number of states are finite, then the system is called finite state machine. For example, a circuit with k registers can be in one of the finite numbers (2^k) of unique states. It is a theoretical model used engineering field to represent and analyze systems that have a finite number of states and transitions between those states. In an FSM, there are a set of states, a set of input symbols or events, a set of transitions between states based on inputs, and a designated initial state. The machine reads input symbols, and based on the current state and input, it transitions to a new state.

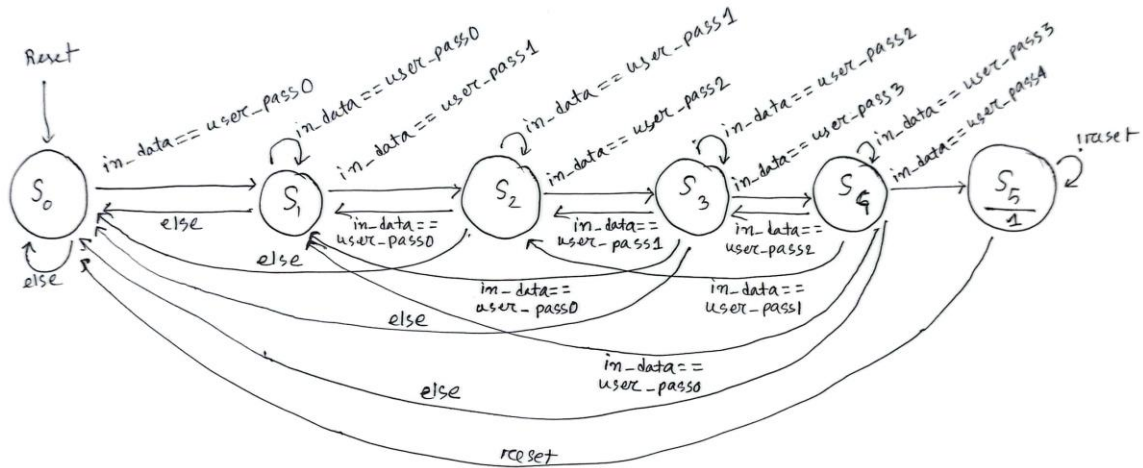
3.2.1 Analysis

In this project we use 5 letter digit password formed between 3 characters 'a', 'b' and 'c'. We can set the password by active password setting mode. After setting password setting mode we can press the button for 'a', 'b' and 'c' to set password. But we can't enter repeated character here. For example, 'abcab', 'acbab' these are valid password. But 'abccb' is not a valid password for our project, because it has 2 repeated character 'c'. For every input of character to set password, led for each corresponding character will light up to confirm our input. After settling password, we can move to password matching mode by switched off the password setting mode switch. Now we can enter password. For every correct enter of password it goes to next state. For every state there has corresponding led. According to the present state corresponding led will light up. If we insert a wrong input, it goes back to the previous correct state of same character. For example, if the password is 'abcab', then if someone first entered 'abc' and then entered 'b' instead of 'a', it goes back to the previous state with character 'b', which is second correct input for this example. After pressing 'reset' button it will again prepare to receive another password. If we want to change the password, again we will press password setting mode switch and enter a new password.

3.3 State Diagram



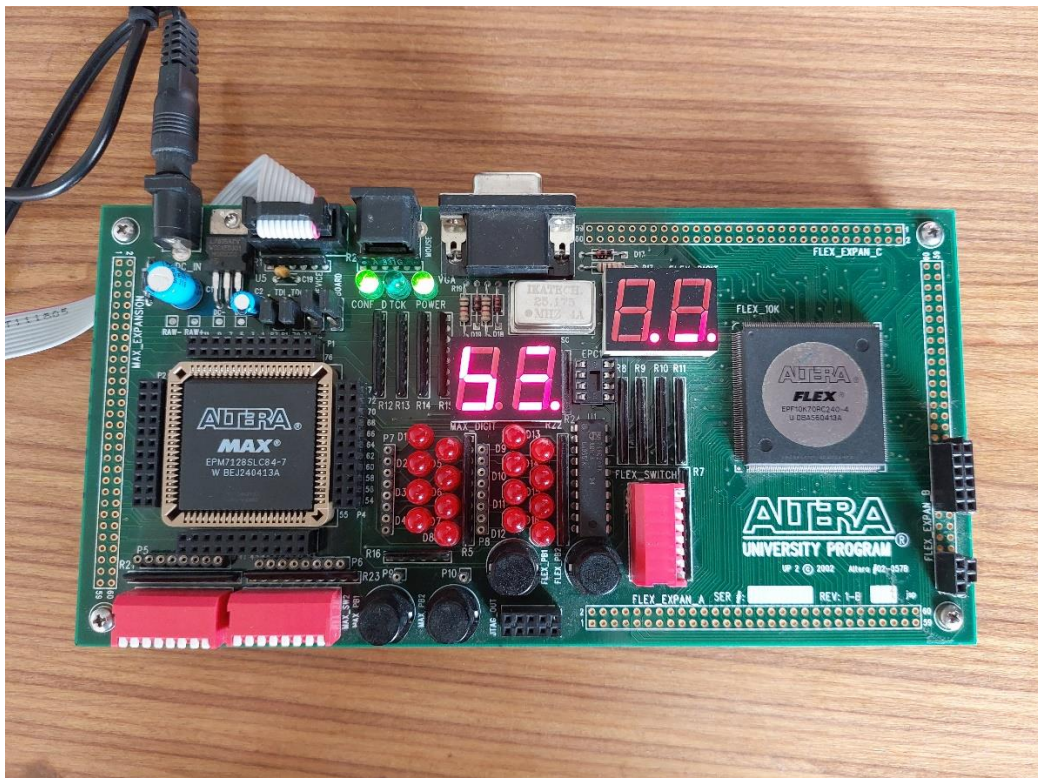
State diagram of password setting mode



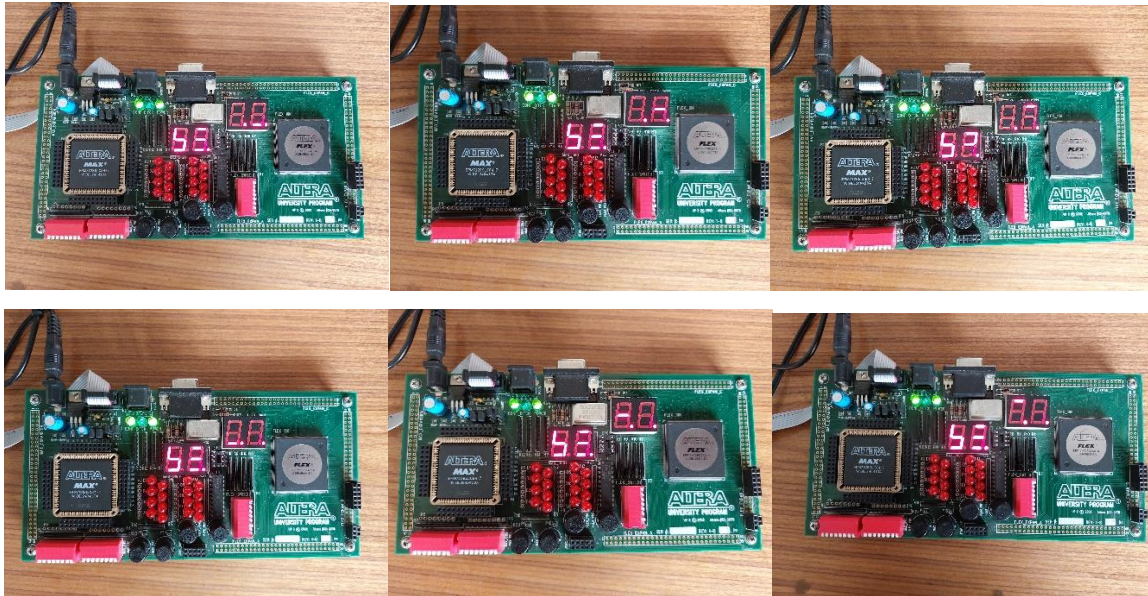
State diagram of password matching mode

3.4 Hardware

We have used UP2 Education Board, Flex10K EPF10K70 device and a USB Blaster connected with a computer as our hardware setup.



UP2 Education Board



Different state with corresponding led in UP2 board

This is UP2 Education Board for our project. Here we have used 6 states with corresponding 6 led. In password setting mode or password matching mode, for each correct input it goes to next state and the corresponding led will turn on and previous state led will turn off.

EPF10K70 Device

The EPF10K70 device is based on SRAM technology. It is available in a 240-pin RQFP package and has 3,744 logic elements (LES) and nine embedded array blocks (EABS). Each LE consists of a four-input LUT, a programmable flipflop, and dedicated signal paths for carry-and-cascade functions. Each EAB provides 2,048 bits of memory which can be used to create RAM, ROM, or first-in first-out (FIFO) functions. EABS can also implement logic functions, such as multipliers, microcontrollers, state machines, and digital signal processing (DSP) functions. With 70,000 typical gates, the EPF10K70 device is ideal for intermediate to advanced digital design courses, including computer architecture, communications, and DSP applications.

3.5 Verilog Code of Firmware

```
module proj1(clk,light,reset,pb0,pb1,pb2,pb3,ledoff,mode);
input  clk,pb0,pb1,pb2,pb3;

input reset,mode;
output reg[7:0] ledoff;
output reg [5:0]light;
parameter [2:0]S0 = 3'b000, S1 =3'b001, S2 =3'b010, S3=3'b011, S4=3'b100, S5=3'b101;
reg [2:0]state, next_state;
parameter [1:0]a=2'b00,b=2'b01,c=2'b10;
reg [1:0]user_password0,user_password1,user_password2,user_password3,user_password4;
reg [1:0]in_data;

//pushbutton setup
always @(posedge clk,negedge reset)
begin
if (!reset)
    in_data=4'b0000;
else begin
if (pb0)
    in_data<=a;
if (pb1)
    in_data<=b;
if(pb2)
    in_data<=c;

end
end

//Cycle of state definition and reset method
always @(posedge clk, negedge reset)
begin
    if(!reset)
        state <=S0;
    else
        state <= next_state;

end

//state transition code for password setting and matching
always @(state,in_data,mode)

    if(mode == 1) begin
        case(state)
            S0: if(pb0 || pb1 || pb2 || pb3)begin
                    if(in_data == user_password0) begin
                        next_state = S1;
                    end end
                    else begin
                        next_state=S0;
                    end

            S1:
                    if(in_data == user_password1) begin
                        next_state = S2;
                    end
                    else if(in_data == user_password0) begin

                        next_state = S1;

                    end

                    else begin
                        next_state=S0;
                    end

            S2:
                    if(in_data == user_password2) begin
                        next_state = S3;
                    end
                    else if(in_data == user_password1) begin
                        next_state = S2;
                    end

end
```

```

        else if(in_data == user_password0) begin
            next_state = S1;
        end
        else begin
            next_state=S0;
        end

S3:
        if(in_data == user_password3) begin
            next_state = S4;
        end
        else if(in_data == user_password2) begin
            next_state = S3;
        end
        else if(in_data == user_password1) begin
            next_state = S2;
        end
        else if(in_data == user_password0) begin
            next_state = S1;
        end
        else begin
            next_state=S0;
        end

S4:
        if(in_data == user_password4) begin
            next_state = S5;
        end
        else if(in_data == user_password3) begin
            next_state = S4;
        end
        else if(in_data == user_password2) begin
            next_state = S3;
        end
        else if(in_data == user_password1) begin
            next_state = S2;
        end
        else if(in_data == user_password0) begin
            next_state = S1;
        end
        else begin
            next_state=S0;
        end

S5:
        begin
            next_state = S5;
        end
    default:
        next_state=S0;
    endcase
end

//password setting if mode ==0
else begin
    case(state)
    S0: if(pb0 || pb1 || pb2 || pb3)begin
        user_password0 = in_data;
        next_state = S1; end
        else
            next_state = S0;

    S1:if(in_data == user_password0)
        next_state =S1;
        else begin
            user_password1 = in_data;
            next_state = S2;end

    S2:if(in_data == user_password1)
        next_state =S2;
        else begin
            user_password2 = in_data;
            next_state = S3;end

    S3:if(in_data == user_password2)
        next_state = S3;
        else begin
            user_password3 = in_data;
            next_state = S4;
        end
    end
end

```

```

        S4:if(in_data == user_password3)
            next_state = S4;
        else begin
            user_password4 = in_data;
            next_state = S5;
        end
    S5:
        begin
            next_state = S5;
        end
    default:
        next_state=S0;
    endcase

end

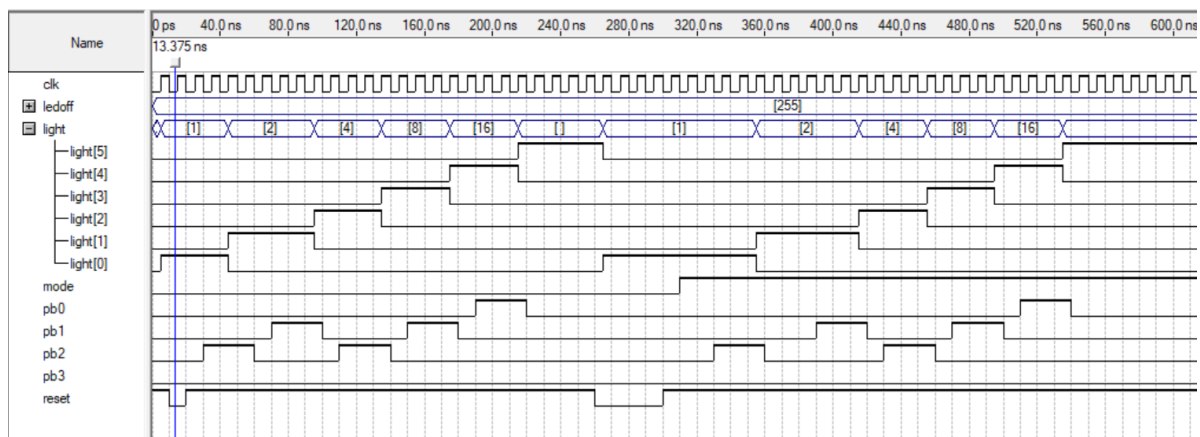
//led lights will turn on according to states
always @(posedge clk)
begin
    light[0] <= ~(state ==S0);
    light[1] <= ~(state == S1);
    light[2] <= ~(state == S2);
    light[3] <= ~(state == S3);
    light[4] <= ~(state == S4);
    light[5] <= ~(state == S5);
    ledoff = 255;
end

endmodule

```

Verilog Code for the main program

3.6 Timing Diagram



Here at first, we are in password setting mode by setting mode value 0 and later we switch to password matching mode by making mode value 1.

4 Implementation

Here we have connected USB Blaster device to connect UP2 Education Board with computer. We have implemented 3 input pushbutton, 1 mode changing switch, 1 reset pushbutton. Three input pushbutton of UP2 Education Board are defined as pin location 41, 40 and 39 respectively. One mode changing switch defined as pinlocation 33. One reset pushbutton defined as pin location 28. Six led L[0] to L[5] are defined as pin location 6, 9, 13, 17, 20, 24 respectively. We have assigned pin as defined location and upload the program to the board by USB Blaster.

5 Design Analysis and Evaluation

5.1 Novelty

A password-based security system implemented using Verilog code and an FPGA board offers several unique advantages and features compared to traditional security methods:

Hardware-Based Security: Unlike software-based solutions, this system leverages the hardware capabilities of an FPGA to process security operations, providing a higher level of security against software-based attacks.

Real-Time Processing: FPGAs are known for their high-speed processing capabilities. This enables real-time authentication and response, enhancing the system's responsiveness and effectiveness.

Customizability: Verilog code allows for highly customizable designs. Developers can tailor the security system to specific requirements and adapt it to different applications or environments.

Low Power Consumption: FPGAs are power-efficient compared to general-purpose processors. This makes them suitable for applications where energy consumption is a concern, such as in embedded systems or IoT devices.

Parallel Processing: FPGAs excel at parallel processing, allowing multiple security checks to be performed simultaneously. This can significantly improve the system's throughput and efficiency.

Compact Form Factor: FPGA boards are typically compact and can be integrated into various hardware designs, making them suitable for space-constrained environments or applications where size matters.

Redundancy and Fault Tolerance: FPGAs can be configured to have redundant components, enhancing system reliability and fault tolerance. This ensures continuous operation even in the presence of hardware failures.

Versatility in Input Methods: While password-based authentication is a key feature, the system can be expanded to support multiple input methods like biometrics, smart cards, or proximity sensors, providing flexibility in authentication modes.

Compliance and Certification: FPGAs and hardware-based security solutions can meet specific industry standards and certifications for security, which is critical in sectors like finance, healthcare, and government.

By combining Verilog code and FPGA technology, a password-based security system gains the benefits of hardware acceleration, customizability, and enhanced security features. This approach is well-suited for applications where robust security, real-time processing, and adaptability to specific requirements are paramount.

5.2 Limitations of Tools

While a password-based security system implemented using Verilog code and an FPGA board offers unique advantages, it also comes with some limitations:

Limited Memory Capacity: FPGAs typically have limited on-chip memory, which can constrain the number of passwords that can be stored and the complexity of the authentication process.

Limited Processing Power: While FPGAs are powerful for specialized tasks, they may not be as capable as general-purpose processors for certain types of computations or complex cryptographic operations.

Programming Complexity: Writing and debugging Verilog code can be complex and time-consuming, requiring specialized knowledge and expertise. This may pose challenges for developers who are not familiar with hardware description languages.

Difficulty in Updating and Maintenance: Modifying or updating the system's functionality may require reprogramming the FPGA, which can be a more involved process compared to software-based systems.

Cost: FPGA boards and associated development tools can be expensive, which may be a barrier for smaller-scale or budget-constrained projects.

Single Point of Failure: If the FPGA board experiences a hardware failure, it could lead to a complete system outage. Implementing redundancy and backup measures may be necessary to mitigate this risk.

Limited User-Friendly Interface: FPGAs are typically not designed for user interaction, which means the system may lack a user-friendly interface for entering passwords or interacting with the security system.

It's important to carefully weigh these limitations against the specific requirements and objectives of the project. Depending on the application, a password-based security system using Verilog code and an FPGA board may be a highly effective solution, but it's crucial to be aware of its constraints.

5.3 Impact Assessment

5.3.1 Assessment of Societal and Cultural Issues

The "Password Based Bank Vault Security System" project carries with it a range of societal

and cultural implications that warrant careful assessment. Firstly, in terms of societal impact, the system's primary function is to provide secure access control to a bank vault, which inherently involves the handling of valuable assets. The project's success or failure directly affects the safety of these assets, making it a matter of significant societal importance. Furthermore, the project's design, including its user interface and accessibility features, plays a role in how effectively it can be adopted and used by individuals from various backgrounds and abilities. Ensuring that the system is user-friendly and inclusive is crucial in this regard.

Culturally, the project's ethical considerations extend to issues of privacy and confidentiality. Given the sensitive nature of bank vault access, it is imperative to respect and safeguard the cultural norms and expectations related to privacy and data security. Cultural nuances should also be considered when developing the system's user interface and instructions, ensuring that they are clear and easily understandable to a diverse user base. Additionally, the project's documentation and support services should be available in multiple languages or adapted to accommodate different cultural contexts.

In summary, the "Password Based Bank Vault Security System" project intersects with societal and cultural dimensions through its impact on asset security, user accessibility, and ethical considerations related to privacy and inclusivity. A holistic approach that prioritizes security, usability, and cultural sensitivity is essential for the project's successful integration into diverse societal and cultural contexts.

5.3.2 Assessment of Health and Safety Issues

The "Password Based Bank Vault Security System" project also necessitates a comprehensive assessment of health and safety issues, as it involves hardware implementation and user interaction. Firstly, from a health perspective, it is crucial to ensure that the hardware components, including the FPGA board and associated peripherals, comply with safety standards to prevent electrical hazards. Adequate ventilation and temperature control should be maintained to prevent overheating and mitigate the risk of fire or damage to electronic components. Users and maintenance personnel should be made aware of potential electrical hazards and take necessary precautions during installation, servicing, or troubleshooting.

On the safety front, the project's user interface, including pushbuttons and LEDs, should be

designed with ergonomics in mind to minimize user strain and discomfort during extended use. Measures should also be in place to prevent physical harm, such as avoiding sharp edges or potential pinch points in the hardware. In cases where the system is used in a physical security context, proper access control measures and safety protocols should be established to prevent unauthorized personnel from tampering with the system.

Additionally, it is essential to consider cybersecurity as a component of safety, as breaches or vulnerabilities in the system can have safety implications for the bank vault and its contents. Robust security measures, including encryption and intrusion detection, should be implemented to protect against unauthorized access or tampering with the system.

In conclusion, the "Password Based Bank Vault Security System" project requires careful consideration of health and safety issues related to electrical hazards, user ergonomics, physical safety, and cybersecurity. By addressing these concerns proactively, the project can ensure that its implementation is both secure and safe for users and the assets it is designed to protect.

5.3.3 Assessment of Legal Issues

The "Password Based Bank Vault Security System" project also demands a thorough assessment of legal issues to ensure compliance with relevant laws and regulations. Firstly, data privacy and protection laws, such as GDPR in Europe or HIPAA in the United States, may apply if the system processes or stores sensitive user information. Compliance with these laws is essential to protect user privacy and avoid legal repercussions.

Additionally, intellectual property rights must be considered. If the project uses third-party software or components, it is crucial to ensure proper licensing and adherence to copyright laws. Furthermore, if the project includes patented technologies or processes, it must respect existing patents and intellectual property rights, seeking legal permission if necessary.

Contractual agreements are another aspect of legal concern. If the "Password Based Bank Vault Security System" is developed in collaboration with others, whether individuals or organizations, clear and legally binding contracts should be in place to define responsibilities, ownership of intellectual property, and dispute resolution mechanisms.

Moreover, compliance with industry-specific regulations may apply. In the financial sector,

for example, there may be specific regulations governing the security and authentication methods used in bank vault access control systems.

Lastly, liability and indemnity considerations are crucial. The project team should have an understanding of their legal liabilities in case of system malfunctions, security breaches, or other unforeseen events and take appropriate measures to protect themselves from legal consequences.

In conclusion, a comprehensive assessment of legal issues, including data privacy, intellectual property, contracts, industry-specific regulations, and liability, is essential for the "Password Based Bank Vault Security System" project to operate within the bounds of the law and mitigate legal risks. Legal counsel and experts in relevant domains should be consulted to ensure full compliance.

5.4 Sustainability and Environmental Impact Evaluation

Evaluating the sustainability and environmental impact of the "Password Based Bank Vault Security System" project is crucial in understanding its implications beyond functionality and security. Here, we'll assess how the project may affect sustainability and the environment:

1. Energy Consumption:

Impact: The operation of the FPGA board and associated hardware components consumes electrical energy.

Evaluation: To assess sustainability, consider measuring the power consumption of the system during operation and standby modes. Evaluate ways to optimize power usage to reduce energy waste.

2. Materials and Resources:

Impact: The production of hardware components, such as FPGA boards, involves the extraction of raw materials and manufacturing processes.

Evaluation: Evaluate the environmental impact of the materials used in the project's hardware components. Consider sustainable sourcing options and recycling practices to minimize resource depletion.

3. E-Waste Generation:

Impact: At the end of the system's lifecycle, hardware components may contribute to electronic waste (e-waste).

Evaluation: Assess the project's plan for disposing of or recycling hardware components to reduce the impact of e-waste. Encourage responsible disposal practices.

4. Manufacturing Processes:

Impact: The manufacturing of FPGA boards and associated hardware involves industrial processes that may emit pollutants.

Evaluation: Consider choosing hardware components from manufacturers with eco-friendly practices and certifications. Evaluate the environmental impact of the manufacturing processes and explore alternatives.

5. User Behavior and Energy Efficiency:

Impact: User behavior can affect energy efficiency, such as leaving the system powered on unnecessarily.

Evaluation: Promote responsible user behavior through awareness campaigns and system features that encourage energy-saving practices, such as automatic standby modes.

6. Project Documentation and Reporting:

Impact: Transparent reporting on the project's environmental impact and sustainability efforts can raise awareness and accountability.

Evaluation: Document the project's sustainability initiatives, challenges faced, and measures taken in the project report. Share this information with stakeholders.

By evaluating these aspects of our project, we can gain a better understanding of its sustainability and environmental impact. Implementing environmentally responsible practices and promoting sustainability can contribute positively to the long-term viability of the "Password Based Bank Vault Security System."

5.5 Ethical Issues

In any project, especially one involving security systems and user data, applying ethical principles is crucial. Below is detailed how ethical principles were applied throughout our "Password Based Bank Vault Security System" project and describe any ethical challenges faced, along with mitigation strategies:

i. User Privacy and Data Security:

- Ethical Principle Applied: Respecting user privacy and ensuring the security of their data is paramount. This includes safeguarding their personal information and passwords.
- Mitigation: To protect user privacy and data, the following measures were taken:
 - User passwords were securely stored in the system, using appropriate encryption and hashing techniques.
 - Access to sensitive data was restricted to authorized personnel only.
 - Users were informed about the importance of keeping their passwords confidential.

ii. Transparency and Informed Consent:

- Ethical Principle Applied: Users should be informed about how their data is being used and should give informed consent for any data processing.
- Mitigation: The project ensured transparency and informed consent by:
 - Providing clear user documentation explaining the system's functionality, including the purpose of password storage.
 - Implementing a user-friendly interface that clearly indicates the current mode (authentication or password setup).
 - Obtaining user consent before allowing them to change their password.

iii. Fairness and Equal Access:

- Ethical Principle Applied: The system should provide equal access and opportunities to all users, without discrimination.
- Mitigation: Fairness and equal access were addressed through:
 - Designing a user-friendly interface that accommodates users with various abilities.
 - Ensuring that the system treats all users equally in terms of authentication and password setup.
 - Avoiding any biases in the system's behavior, such as favoring certain users.

iv. Accountability and Responsibility:

- Ethical Principle Applied: Project developers are accountable for the system's functionality and security.
- Mitigation: Accountability and responsibility were addressed by:
 - Conducting thorough testing and quality assurance to identify and resolve potential security vulnerabilities.
 - Appointing a project administrator or security officer responsible for overseeing system integrity and user data protection.
 - Establishing procedures for responding to security incidents or data breaches promptly.

v. User Education and Awareness:

- Ethical Principle Applied: Users should be educated about the system's use, security best practices, and potential risks.
- Mitigation: To promote user education and awareness:
 - Comprehensive user documentation was provided, explaining how to use the system securely and effectively.
 - Users were encouraged to keep their passwords confidential and change them regularly.
 - System administrators were available to answer user questions and provide assistance.

vi. Ethical Challenges and Mitigation:

- Challenge 1: Data Breach Risk: Storing user passwords creates a risk of data breaches if not handled securely.
 - Mitigation: Strong encryption and hashing techniques were employed to protect stored passwords. Regular security audits were conducted to identify and address vulnerabilities.
- Challenge 2: Password Disclosure: Users might inadvertently disclose their passwords to others.
 - Mitigation: The system encouraged users to keep their passwords confidential and did not store them in plaintext. Password setup procedures were designed to be user-friendly and minimize the chance of password leakage.

- Challenge 3: Discrimination: Unintentional biases in the system's behavior could lead to discrimination.

- Mitigation: Thorough testing was conducted to ensure that the system treated all users equally. Developers were vigilant about avoiding any biases in system behavior.

By applying ethical principles and addressing potential challenges, your project aimed to create a secure and fair "Password Based Bank Vault Security System" that respects user privacy and data security while providing equal access and promoting responsible use. Regular monitoring and ongoing ethical considerations are essential to maintaining these principles throughout the system's lifecycle.

6 Reflection on Individual and Team work

6.1 Mode of Team Work

We have divided the total work in two segments. Where two of us done the most of the verilog code and rest of us done the integration of FPGA board to our laptop. Besides as a team we all try to build up the state diagram which is the backbone of our project. Like the project, we all paid same effort individually to prepare the slides, video and report.

6.2 Log Book of Project Implementation

Date	Milestone Achieved	Individual Role	Team Role	Comments
04-07-2023	Project Kickoff and Scope Definition	Nilotpaul: Defined project architecture	Team Discussion: Defined objectives	Project officially launched, roles assigned.
28-07-2023	Verilog Code Development	Swapnil: Developed core Verilog code	Enam Assisted with hardware integration	Initial Verilog code for the system implemented.
15-08-2023	FPGA Hardware Integration	Enam: Integrated Verilog with FPGA	Conducted initial testing	FPGA board successfully integrated with Verilog code.
22-08-2023	Multilingual Testing and User Feedback	Antor:Conducted user testing	Nilotpaul: Assisted with user feedback	Positive feedback received; minor adjustments made.
29-08-2023	Final System Integration and Testing	Nilotpaul: Oversaw FPGA integration	Swapnil: Conducted final Verilog testing	Comprehensive system testing and debugging completed.
05-09-2023	Documentation and Ethical Considerations	Antor: Documented project details	Discussion on ethical principles	Project documentation and ethical considerations added.
11-09-2023	Project Presentation and Review	Antor: Prepared project presentation	Team Presentation: Reviewed project	Project presented successfully, readiness for deployment.

7 Communication

Our team presents the "Password Bank Vault," a revolutionary security system. This section highlights its user-friendly design, simplifying password setup and ensuring secure access to your bank vault.

7.1 Executive Summary

Revolutionizing Security: Introducing the "Password Bank Vault"

In a breakthrough project, our team proudly presents the "Password Bank Vault," a cutting-edge security system that ensures your assets are safeguarded like never before. Simple to use and highly secure, our system employs state-of-the-art technology to protect your valuables. With user-friendly features and top-notch encryption, you're in control. Set up your unique password effortlessly, and access your bank vault with confidence. Your security, our priority. Join us in the future of security. Experience the "Password Bank Vault" today!

7.2 User Manual

Welcome to the user guide for the Password Based Bank Vault Security System. This guide will walk you through the process of using the system to access a bank vault securely. Please follow these steps carefully to ensure smooth operation.

Step 1: Power On the System

1. Ensure that the FPGA board is connected to a power source and is powered on.
2. Wait for the board to initialize and the system to start.

Step 2: Understand the System Interface

The system interface consists of pushbuttons, LEDs, and a display. Here's what each component does:

- Pushbuttons (PB0, PB1, PB2, PB3): These buttons are used to enter digits of your password 'a', 'b', 'c' and 'd' to navigate through the system.
- LEDs (Light): These LEDs provide feedback about the system's current state.
- Display (if available): In an enhanced version of the system, you may have a display to show messages or instructions.

Step 3: Select Operation Mode

The system has two main modes: Authentication Mode (Mode 1) and Password Setup Mode (Mode 0).

- Authentication Mode (Mode 1): This is the default mode where you can enter your password to access the bank vault.
- Password Setup Mode (Mode 0): In this mode, you can set or update your password. This mode is used for initial setup or changing your password.

Step 4: Entering Password (Authentication Mode - Mode 1)

1. Ensure that the system is in Authentication Mode (Mode 1). You may need to switch modes using the mode selection option if available.
2. Observe the LEDs (Light) to confirm that the system is in the "Idle" state (S0).
3. Enter your password digit by digit using PB0, PB1, PB2, and PB3.
 - Press PB0, PB1, PB2, or PB3 to enter digits (a to d) respectively.
 - Press the buttons in the correct sequence to enter your full password.
4. The system will process your input and transition through states.
5. Observe the LEDs to see if access is granted (S5) or if the authentication fails.

Step 5: Password Setup (Password Setup Mode - Mode 0)

1. To set or update your password, switch the system to Password Setup Mode (Mode 0) if available.
2. The system may prompt you to enter the current password if you are updating it.
3. Follow the same procedure as in Step 4 to enter your new password digit by digit.
4. The system will update your password accordingly.
5. Observe the LEDs to see your input.

Step 6: System Feedback

- The LEDs provide feedback on the system's state:
- Illuminated LEDs indicate the current state.
- Access Granted: All LEDs may be off (depending on implementation).
- Authentication Failed: LEDs will indicate the failed state.

Step 7: Access Vault (If Granted)

- If access is granted (S5), follow the instructions provided by the system to access the bank vault.

Step 8: Power Off the System

1. When you have finished using the system, power it off.
2. Disconnect the FPGA board from the power source.

Additional Tips:

- Keep your password confidential and avoid sharing it with others.
- If you forget your password, consult the system administrator for password reset procedures.
- If you encounter any issues or have questions, refer to the project documentation or seek assistance from the system administrator.

8 Future Work

Future enhancements for the project may include:

- Integration with additional security features (biometric authentication, alarms).
- Improved user interface and display.
- Remote access and monitoring capabilities.
- Enhanced security protocols and encryption.

9 References

- *“Design of a Bank Vault Security System with Password, Thermal & Physical Interrupt Alarm”* by Sayemul Islam and Md. [Saiduzzaman](#)
- *“Intelligent RFID biometric enabled dual security lock in the banking sector”* by S. [Ramani et al](#)
- *“Password Based Lock System using Verilog HDL & Proteus Simulation”* by [EEE 304](#)
- *“Bank Vault Security System with Password and GSM Technology”* by R. Sathish Kumar and K. Srinivasan .
- *“A Secure Bank Locker System using RFID and GSM Technology”* by R. Sathish Kumar and K. Srinivasan .
- *“Design and Implementation of a Bank Security System Based on Fingerprint Recognition”* by Xuefeng Liu et al. .
- *“A Secure Bank Locker System using RFID and GSM Technology”* by R. Sathish Kumar and K. Srinivasan .
- *“A Secure Bank Locker System using RFID and GSM Technology”* by R. Sathish Kumar and K. Srinivasan .